

CCL-Diff: Representation-Consistent Diffusion with Intrinsic Contrastive Learning for Recommender Systems

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Abstract

Diffusion models have gained traction in recommender systems for learning user preferences via generative denoising. However, they often overlook an essential aspect: the semantic consistency of representations across diffusion timesteps, limiting their robustness and generalizability. To address this, we propose CCL-Diff, a novel diffusion-based recommendation framework enhanced by Cross-Denoising Contrastive Learning (CCL). CCL exploits the intrinsic multi-view nature of the diffusion process, treating intermediate noisy states as semantically consistent views. By aligning denoised embeddings across these views, it enforces noise-invariant user representations, enhancing the model's discriminative power. Unlike traditional graph contrastive learning, which relies on manual augmentations, CCL derives contrastive signals directly from the denoising trajectory, eliminating the need for external perturbations. CCL is a lightweight, plug-and-play component that integrates effortlessly into diffusion models. We also introduce a cross-attention denoising decoder to adaptively fuse features with timestep-specific embeddings, effectively capturing the temporal dynamics of user preferences. Experimental results on three datasets demonstrate that CCL-Diff consistently outperforms state-of-the-art models. Moreover, applying CCL as an auxiliary objective boosts the performance of diffusion models, highlighting its broad applicability.

CCS Concepts

• Information systems → Recommender systems.

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Keywords

Diffusion Models, Cross-Denoising Contrastive Learning, Recommender Systems, Representation Consistency

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1 Introduction

Graph Convolutional Networks (GCNs) have become a cornerstone of modern recommender systems. Methods like LightGCN [7] achieve impressive performance by neighbor aggregation on the user-item interaction graph. Further advances incorporate Graph Contrastive Learning (GCL) [6, 18, 35, 42] to enhance representation learning by enforcing invariance to handcrafted perturbations (e.g., node/edge dropout). However, these approaches primarily operate on static graphs with augmentations, struggling to capture the dynamic nature of user preferences. Diffusion models (DMs) [28, 44] provide a promising alternative by reframing recommendation as a generative denoising process. Recent works such as DiffRec [33], CF-Diff [10], and G-Diff [3] demonstrate that progressively injecting and removing noise can yield robust user representations.

Nevertheless, a critical property of the diffusion process in recommendation remains largely unexplored. As illustrated in Figure 1 (a), unlike applications in computer vision [2, 4, 8, 9, 23, 29, 34] or natural language processing [1, 16, 24] that employ hundreds of noise steps, DMs for recommendation are typically limited to shallow depths (tens of steps) due to the sparsity of user-item graphs. Under such shallow diffusion, the intermediate noisy representations remain strong correlations with the original data, inherently forming multiple, partially corrupted yet semantically consistent views of user preference, a unique characteristic of the recommendation context. Surprisingly, existing DMs [3, 10, 33] neglect this intrinsic multi-view structure and the semantic overlap across timesteps. They focus predominantly on reconstructing observed interactions, failing to explicitly enforce representation consistency across the

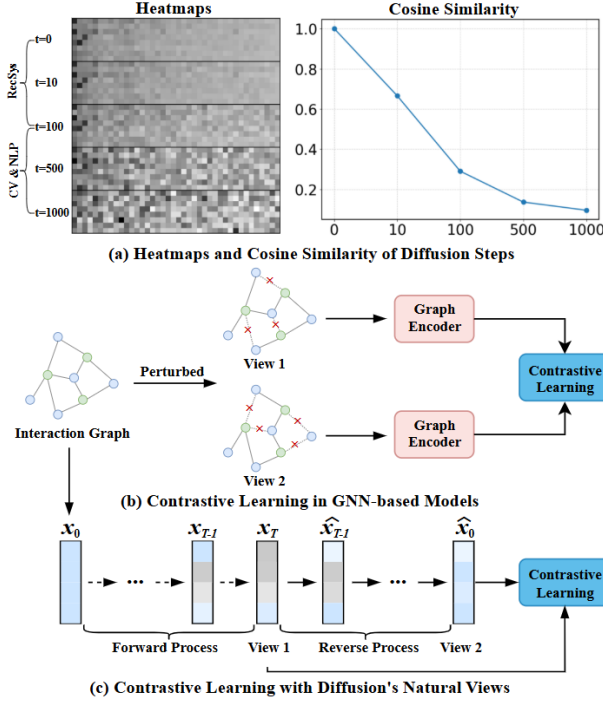


Figure 1: Conceptual comparison of GCL and CCL. (a) Shallow diffusion in recommendation preserves semantic consistency, creating natural intermediate views. (b) GCL uses manual perturbations (e.g., edge/node dropout) to generate contrastive views. (c) CCL exploits the multi-view structure of diffusion, aligning denoised embeddings from different timesteps without the need for external augmentation.

denoising trajectory. This oversight leads to two key limitations: (1) increased susceptibility to overfitting on sparse data, and (2) incoherent representations across timesteps, ultimately impairing model robustness and generalization.

To address these issues, we propose CCL-Diff, a novel framework that introduces Cross-Denoising Contrastive Learning (CCL) into the diffusion process. CCL is designed to explicitly leverage the intermediate noisy states as naturally occurring, semantically consistent views. During training, it enforces agreement (e.g., via contrastive loss) between the denoised embeddings derived from these different noisy states across timesteps. This intrinsic contrast signal acts as a powerful regularizer, promoting a locally smooth and global stable latent space. It guides the model to learn more robust and generalizable user preference representations, effectively compensating for the limitation of shallow diffusion depths typically used in recommendation. As illustrated in Figure 1 (b) and (c), unlike conventional GCL methods that rely on manually crafted perturbations such as node or edge dropout to generate contrastive views, CCL operates entirely within the native diffusion pipeline. It seamlessly deriving contrastive signals from the inherent denoising process, without introducing additional augmentation or architectural modifications. As a result, CCL serves as a plug-and-play module that can be integrated into diffusion-based frameworks

with negligible computational overhead. Furthermore, we introduce a cross-attention denoising decoder (A-MLP) to better capture fine-grained temporal dynamics during denoising. The A-MLP module dynamically fuses timestep-specific diffusion embeddings with user features via cross-attention, enabling the model to adaptively focus on evolving preference signals throughout the diffusion trajectory.

Our principal contributions are summarized as follows:

- We propose CCL-Diff, a novel diffusion-based recommendation framework that incorporates intrinsic contrastive learning into the denoising process to enforce representation consistency across timesteps.
- We introduce the Cross-denoising Contrastive Learning (CCL) objective, which leverages intermediate noisy states as natural contrastive views. This objective provides a built-in regularization effect without data augmentation, and we show it can be applied as a plug-and-play module to boost existing diffusion models by up to 20.1% in Recall@10.
- We design a cross-attention denoising decoder (A-MLP) that adaptively fuses timestep-aware embeddings with user features to capture temporal preference dynamics.
- We conduct extensive experiments on three public benchmarks, showing that CCL-Diff consistently outperforms state-of-the-art models, improving R@10 by up to 13.1%, and generalizes robustly under sparse data settings.

2 Related Work

2.1 GCN-Based Recommendation

Graph Neural Networks (GNNs) [5, 11, 36] effectively capture high-order connectivity for recommendation. Notably, LightGCN [7] establishes a strong baseline by removing nonlinearities to simplify the architecture. To mitigate data sparsity, Graph Contrastive Learning (GCL) [12, 19] introduces self-supervised signals via stochastic dropout (SGL [35]) or noise injection and prototype alignment (XSimGCL [42], NCL [18]). However, these methods primarily rely on static structures and manual augmentations, lacking the generative capacity for modeling complex user preference transitions.

2.2 Generative-Based Recommendation

2.2.1 VAE-Based Methods. Variational Autoencoders (VAEs) [21, 39, 40, 43, 45] are widely used to learn representations from implicit feedback. Early denoising frameworks like DAE [37] and CVAE [13] evolved into models like Mult-VAE [17] and RecVAE [25], which incorporate multinomial likelihoods for better interaction modeling. Despite their utility, VAE-based approaches often struggle to balance simplicity with expressiveness for intricate behaviors.

2.2.2 Diffusion-Based Methods. Diffusion Models (DMs) [15, 27, 41, 46] have emerged as a powerful paradigm by modeling recommendation as a progressive denoising process. CODIGEM [30] first adapted DDPMs [8] for collaborative filtering. Recent studies have extended this to diverse settings: DiffRec [33] and G-Diff [3] incorporate sequential and graph structural signals, respectively, while CF-Diff [10] leverages cross-attention to integrate multi-hop neighbor information. Nevertheless, existing diffusion methods lack explicit mechanisms to enforce representation consistency across timesteps, limiting their generalizability in sparse scenarios.

3 Preliminary

DMs have demonstrated powerful generative capabilities across various domains. A typical DM consists of a forward (noising) process and a reverse (denoising) process [8, 26].

3.1 Forward process

Given a data sample $\mathbf{x}_0 \sim q(\mathbf{x}_0)$, the forward process constructs a sequence of latent variables $\mathbf{x}_{1:T}$ through a Markov chain by progressively adding Gaussian noises over T steps. The transition from $\mathbf{x}_{t-1}$ to $\mathbf{x}_t$ is define as:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I}), \quad (1)$$

where $\alpha_t = 1 - \beta_t$, $t \in \{1, \dots, T\}$, $\mathcal{N}$ denotes the Gaussian distribution, and $\beta_t \in (0, 1)$ controls the noise scales at step t . As $T \rightarrow \infty$, $\mathbf{x}_T$ converges to a standard Gaussian distribution [8].

3.2 Reverse process

The reverse process aims to recover $\mathbf{x}_{t-1}$ from $\mathbf{x}_t$ by iteratively removing the noise added during the forward process. This is formulated as:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)), \quad (2)$$

where $\boldsymbol{\mu}_\theta(\mathbf{x}_t, t)$ and $\Sigma_\theta(\mathbf{x}_t, t)$ are parameterized by a neural network with parameters θ . The objective is to approximate the true reverse transition distribution, enabling the reconstruction of $\mathbf{x}_0$ from $\mathbf{x}_T$.

3.3 Training

DMs are trained by maximizing the evidence lower bound (ELBO) of the log-likelihood of the observed data $\mathbf{x}_0$:

$$\begin{aligned} \log p(\mathbf{x}_0) &= \log \int p(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} = \log \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\frac{p(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] \\ &= \underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]}_{\text{Reconstruction}} - \underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{\text{Prior}} \\ &\quad - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]}_{\text{Denoising}}, \end{aligned} \quad (3)$$

where the *reconstruction* term measures the recovery accuracy of $\mathbf{x}_0$, the *prior* term is a constant with no trainable parameters and can be omitted during optimization, and the *denoising* term regularizes $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ to match the tractable posterior $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ [20]. Following the research [8], the denoising matching terms can be simplified into a mean-squared error objective:

$$\sum_{t=2}^T \mathbb{E}_{t, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|_2^2], \quad (4)$$

where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and $\epsilon_\theta(\mathbf{x}_t, t)$ is a neural network (e.g., U-Net [8]) that predicts the noises added in the forward process [20].

3.4 Inference

Once trained, the model generates samples by first drawing a noise vector $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and then iteratively applying the learned reverse process $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ to produce the sequence $\mathbf{x}_T \rightarrow \mathbf{x}_{T-1} \rightarrow \dots \rightarrow \mathbf{x}_0$. To support controllable generation, the reverse process

can be conditioned on additional inputs, such as user or item embeddings in recommendation scenarios, enabling the synthesis of personalized interaction patterns [14, 23].

4 Methodology

As shown in Figure 2, CCL-Diff incorporates lightweight graph augmentation (node/edge dropout) prior to diffusion to enhance structural robustness. Crucially, this step is independent of the CCL objective, which derives contrastive signals exclusively from diffusion-induced noise rather than constructed views. During the reverse process, the A-MLP module progressively reconstructs interactions for recommendation, while CCL operates in parallel to align representations across timesteps.

4.1 Forward Process

Let $\mathbf{x} \in \mathbb{R}^{|\mathcal{U}| \times |\mathcal{I}|}$ denote the user-item interaction matrix, and $\mathbf{x}_u \in \mathbb{R}^{|\mathcal{I}|}$ the interaction vector of user u . We first apply graph augmentation via node and edge dropout with probability ρ to obtain the perturbed input $\mathbf{x}_0$. Specifically, node dropout masks a subset of users:

$$\mathbf{x}^{\text{node}} = \mathbf{x} \odot \mathbf{m}^{\text{node}}, \quad \mathbf{m}^{\text{node}} \sim \text{Bernoulli}(1 - \rho)^{|\mathcal{U}|}, \quad (5)$$

and edge dropout further corrupts the observed interactions:

$$\mathbf{x}_u^{\text{aug}} = \mathbf{x}_u^{\text{node}} \odot \mathbf{m}^{\text{edge}}, \quad \mathbf{m}^{\text{edge}} \sim \text{Bernoulli}(1 - \rho)^{|\mathcal{I}|}. \quad (6)$$

We denote the ground-truth interaction vector as $\mathbf{x}_G = \mathbf{x}^1$ and the augmented input as $\mathbf{x}_0 = \mathbf{x}_u^{\text{aug}}$. The forward diffusion process is then defined as a Markov chain:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I}), \quad (7)$$

where $\beta_t \in (0, 1)$ controls the variance of the Gaussian noise at step t .

Using the reparameterization trick and the additivity of Gaussian noise [8, 20], we can directly express $\mathbf{x}_t$ in terms of $\mathbf{x}_0$ as:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad (8)$$

where $\alpha_t = 1 - \beta_t$, and $\bar{\alpha}_t = \prod_{l=1}^t \alpha_l$. This allows the sample $\mathbf{x}_t$ to be rewritten as $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$, with $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. We adopt a linear noise schedule for $1 - \bar{\alpha}_t$ to regulate the noise intensity:

$$1 - \bar{\alpha}_t = s \cdot \left[\alpha_{\min} + \frac{t-1}{T-1} (\alpha_{\max} - \alpha_{\min}) \right], \quad t \in \{1, \dots, T\}, \quad (9)$$

where $s \in [0, 1]$ is a scale hyperparameter, and $\alpha_{\min} < \alpha_{\max} \in (0, 1)$ define the lower and upper bounds of the added noise.

4.2 Reverse Process

The reverse process aims to reconstruct the user interaction vector by iteratively denoising from the initial state $\mathbf{x}_T$ through the learned denoising transition:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)), \quad (10)$$

where $\boldsymbol{\mu}_\theta(\mathbf{x}_t, t)$ and $\Sigma_\theta(\mathbf{x}_t, t)$ are parameterized by a neural network with weights θ .

Existing approaches [17, 33] typically implement the denoising network $\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)$ using a MLP that takes the noisy representation $\mathbf{x}_t$ and the timestep t as input. However, such designs often underutilize the temporal structure inherent in the diffusion trajectory.

¹For notation brevity, we omit the user subscript u in $\mathbf{x}_G$.

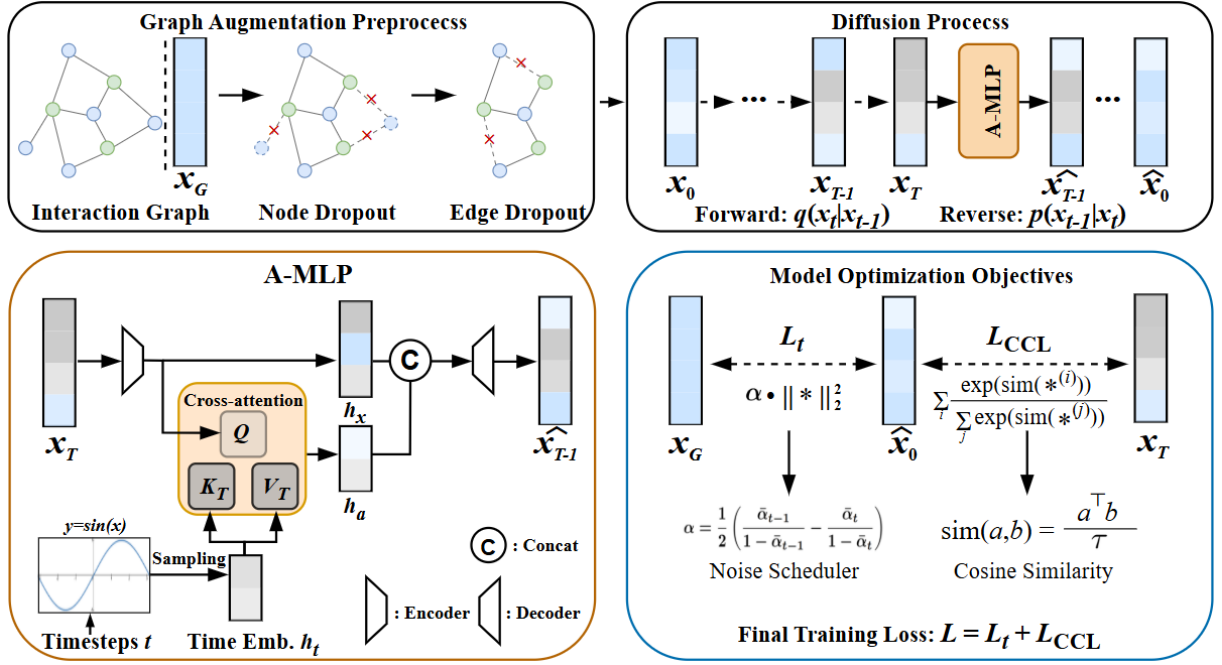


Figure 2: Overall architecture of CCL-Diff. The framework first perturbs the input graph via stochastic node and edge dropout. The reverse denoising process, powered by the A-MLP decoder, then reconstructs the original user-item interactions. The model is jointly trained with a denoising loss $\mathcal{L}_t$ and the proposed CCL objective $\mathcal{L}_{CCL}$, which enforces representation consistency across diffusion timesteps.

To better capture temporal dynamics, we propose an A-MLP, which incorporates a cross-attention mechanism to inject step-specific inductive biases into the denoising process.

Formally, let $\mathbf{x}_t \in \mathbb{R}^{N \times |I|}$ denote a batch of noisy interaction vectors. The A-MLP operates as follows:

- **Step 1: Timestep Embedding.** We encode the diffusion step t into a continuous representation using sinusoidal embedding:
$$\mathbf{e}_t^{(2i)} = \cos(t/\omega_i), \quad \mathbf{e}_t^{(2i+1)} = \sin(t/\omega_i), \quad (11)$$
where $\omega_i = 10000^{2i/d_e}$. A linear projection is then applied to obtain a learnable timestep representation $\mathbf{h}_t \in \mathbb{R}^{d_e}$.
- **Step 2: MLP Encoding.** The input $\mathbf{x}_t$ is optionally normalized, passed through a dropout layer, and then fed into a stack of fully connected layers (two layers in our implementation). This produces an intermediate feature $\mathbf{h}_x \in \mathbb{R}^{N \times d}$.
- **Step 3: Cross-Attention.** To integrate temporal context, $\mathbf{h}_x$ is projected into $\mathbf{z}_x \in \mathbb{R}^{N \times d_e}$ via a linear layer. A multi-head attention mechanism with $H = 4$ heads is then applied, where the query comes from $\mathbf{z}_x$ and both keys and values come from $\mathbf{h}_t$:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_e/H}}\right)V. \quad (12)$$

This produces timestep-aware features $\mathbf{h}_a \in \mathbb{R}^{N \times d_e}$.

- **Step 4: Fusion and Output.** The features $\mathbf{h}_x$ and $\mathbf{h}_a$ are concatenated as $[\mathbf{h}_x; \mathbf{h}_a] \in \mathbb{R}^{N \times (d+d_e)}$, and then passed through

a stack of output MLP layers (two layers in our implementation) to generate the final denoised prediction $\hat{\mathbf{x}}_0 \in \mathbb{R}^{N \times |I|}$.

4.3 Cross-Denoising Contrastive Learning (CCL)

CCL enhances representation robustness by enforcing consistency between two views derived intrinsically from the diffusion process: the fully noised state $\mathbf{x}_T$ (View 1) and the denoised output $\hat{\mathbf{x}}_0$ (View 2). We deliberately select $\mathbf{x}_T$ over intermediate states $\mathbf{x}_t$ to maximize semantic disparity, as analyzed in Section *Theoretical Analysis of View 1 Selection*. Notably, CCL requires no external data augmentation; it exploits the inherent stochasticity of the diffusion process to construct contrastive views.

The cosine similarity with temperature τ is defined as:

$$\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}^\top \mathbf{b}}{\tau}, \quad \text{where } \mathbf{a}, \mathbf{b} \in \mathbb{R}^d. \quad (13)$$

We then adopt the InfoNCE loss to align each positive pair $(\mathbf{x}_T^{(i)}, \hat{\mathbf{x}}_0^{(i)})$ while repelling negative pairs within the batch:

$$\mathcal{L}_{CCL} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(\mathbf{x}_T^{(i)}, \hat{\mathbf{x}}_0^{(i)}))}{\sum_{j=1}^N \exp(\text{sim}(\mathbf{x}_T^{(i)}, \hat{\mathbf{x}}_0^{(j)}))}. \quad (14)$$

All representations are ℓ_2 -normalized prior to similarity computation.

Algorithm 1 Training**Input:** all users' interactions X and randomly initialized θ .

```

1: repeat
2:   Sample a batch of users' interactions  $x \subset X$ ;
3:   Perform graph augmentation on  $x$  in Eq. (5) and (6).
4:   for all  $x_0 \in x$  do
5:     Sample  $t \sim \mathcal{U}(1, T)$  or  $t \sim p_t$ ,  $\epsilon \sim \mathcal{N}(\mathbf{0}, I)$ ;
6:     Compute  $x_t$  given  $x_0$ ,  $t$ , and  $\epsilon$  in Eq. (8);
7:     Use A-MLP to predict  $\hat{x}_0 = \hat{x}_\theta(x_t, t)$ ;
8:     Compute contrastive loss  $\mathcal{L}_{\text{CCL}}$  via Eq. (14);
9:     Compute  $\mathcal{L}_t$  with coefficient  $\alpha$  via Eq. (15);
10:    Update  $\theta$  via gradient descent on total loss  $\mathcal{L}$ ;
11:   end for
12: until converged
Output: optimized  $\theta$ .
```

4.4 Training and Inference

The model parameters θ are learned by minimizing a diffusion reconstruction loss combined with the CCL objective. Following [33], we optimize the evidence lower bound (ELBO) via a simplified mean-squared error objective:

$$\mathcal{L}_t = \mathbb{E}_{q(x_t|x_G)} [\| \hat{x}_\theta(x_t, t) - x_G \|_2^2], \quad (15)$$

where $\hat{x}_\theta(x_t, t)$ denotes the model's reconstruction of the ground-truth interaction vector x_G from the noisy input x_t at step t . For $t > 1$, the loss is weighted by a coefficient α defined as:

$$\alpha = \frac{1}{2} \left(\frac{\bar{\alpha}_{t-1}}{1 - \bar{\alpha}_{t-1}} - \frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t} \right). \quad (16)$$

The total training loss is then given by:

$$\mathcal{L} = \mathcal{L}_t + \mathcal{L}_{\text{CCL}}. \quad (17)$$

The overall training procedure is summarized in Algorithm 1.

During inference, CCL-Diff first corrupts the input x_0 through T' forward steps to obtain $\hat{x}_T = x_{T'}$, and then executes the reverse denoising process for T steps: $\hat{x}_T \rightarrow \hat{x}_{T-1} \rightarrow \dots \rightarrow \hat{x}_0$. We set $T' < T$ for two reasons: (1) real-world user interactions often contain noise (e.g., false positives/negatives) [31, 32], and (2) it helps preserve personalized signals by reducing the intensity of added noise. The final output $\hat{x}_0$ is used for item ranking and top- K recommendation.

4.5 Theoretical Analysis of View 1 Selection

We present an information-theoretic analysis to justify our selection of x_T and $\hat{x}_0$ as contrastive views in CCL. This analysis demonstrates that under the shallow diffusion settings typical in recommendation systems, this specific pairing maximizes the mutual information gain, thereby enhancing the quality of learned representations. Empirical validation of this theoretical insight is provided in Section 5.2.2.

Mutual Information Decay. Let z_u denote the latent user interest vector and x_t the perturbed interaction vector at timestep t . The forward diffusion process progressively injects noise, reducing the mutual information between $I(x_t; z_u)$ as t increases. Formally,

this relationship is expressed as:

$$I(x_t; z_u) = -\frac{d}{2} \log(1 - \bar{\alpha}_t) + C, \quad (18)$$

where C is a constant and $\bar{\alpha}_t \rightarrow 0$ as $t \rightarrow T$. Consequently, the mutual information reaches its minimum at $t = T$.

This derivation follows from the forward process definition $x_t = \sqrt{\bar{\alpha}_t} z_u + \sqrt{1 - \bar{\alpha}_t} \epsilon$, which gives the conditional distribution:

$$p(x_t|z_u) = \mathcal{N}(\sqrt{\bar{\alpha}_t} z_u, (1 - \bar{\alpha}_t)I). \quad (19)$$

The mutual information can then be expressed as:

$$I(x_t; z_u) = h(x_t) - h(x_t|z_u) \propto -\log(1 - \bar{\alpha}_t). \quad (20)$$

Optimality of View Selection. The pairing of x_T (the maximally noisy input) and $\hat{x}_0$ (the denoised output) enables maximal information gain during contrastive learning. Under shallow diffusion conditions where $T \ll 1000$, we establish the lower bound:

$$I(\hat{x}_0; z_u) \geq I(x_T; \hat{x}_0) + I(z_u; \hat{x}_0|x_T) - \epsilon(T), \quad (21)$$

where $\epsilon(T) \propto \frac{1}{T}$ represents the diffusion gap. This bound derives from the chain rule of mutual information:

$$I(z_u; x_T, \hat{x}_0) = I(z_u; x_T) + I(z_u; \hat{x}_0|x_T), \quad (22)$$

coupled with the data processing inequality:

$$I(z_u; \hat{x}_0) \leq I(z_u; x_T, \hat{x}_0). \quad (23)$$

Since $I(z_u; x_T) \approx 0$ when $t = T$, maximizing $I(x_T; \hat{x}_0)$ directly tightens the bound for $I(\hat{x}_0; z_u)$.

Conclusion. This analysis provides a rigorous foundation for selecting x_T and $\hat{x}_0$ as contrastive views. Because x_T contains minimal information about user interests while $\hat{x}_0$ is semantically rich, contrasting these two states maximizes the conditional information gain:

$$I(z_u; \hat{x}_0|x_T). \quad (24)$$

This gain would be substantially smaller if intermediate state x_t (where $x < T$) were used indeed, due to their higher residual mutual information with the latent user representation.

5 Experiments

We conduct extensive experiments on three real-world datasets to address the following research questions:

RQ1: How does CCL-Diff perform compared to SOTA recommendation models?

RQ2: How does key hyperparameters influence the performance of CCL-Diff?

RQ3: What is the contribution of each component in CCL-Diff to the overall performance?

RQ4: Can CCL be effectively integrated into existing DMs to enhance their performance?

RQ5: How does the computational efficiency of CCL-Diff compare with baseline DMs?

5.1 Experimental Setup

5.1.1 Datasets. We evaluate our method on three public benchmarks: (1) **Amazon book**² is a subset of the Amazon review dataset containing rich user-book interactions. (2) **Anime**³ is a large-scale

²<https://cseweb.ucsd.edu/~jmcauley/datasets/amazon/links.html>

³<https://www.kaggle.com/datasets/CooperUnion/anime-recommendations-database>

Table 1: Statistics of the evaluation datasets.

Datasets	#User	#Item	#Review	Density (%)
Book	108,822	94,949	3,146,256	0.0304
Anime	54,199	6,969	7,634,582	2.0216
Yelp	54,574	34,395	1,402,736	0.0747

anime recommendation dataset collected from MyAnimeList. (3) **Yelp**⁴ is a representative business dataset containing user reviews for different restaurants and services. Following common practice, we split each user’s interactions chronologically: the first 70% for training, the next 10% for validation (hyperparameter turning), and the remaining 20% for testing. Key statistics of the datasets are summarized in Table 1.

5.1.2 Evaluation. We adopt the full-ranking evaluation protocol [7], ranking all non-interacted items for each user. Performance is measured using two standard metrics: Recall@K (R@K) and Normalized Discounted Cumulative Gain@K (N@K), with $K \in \{10, 20\}$.

5.1.3 Baselines. We compare CCL-Diff against eleven strong baselines, including GNN-, AE-, and diffusion-based methods: MF [22], LightGCN [7], SGL [35], NCL [18], XSimGCL [42], CDAE [38], Multi-DAE [17], MultiVAE [17], DiffRec [33], G-Diff [3], and CF-Diff [10].

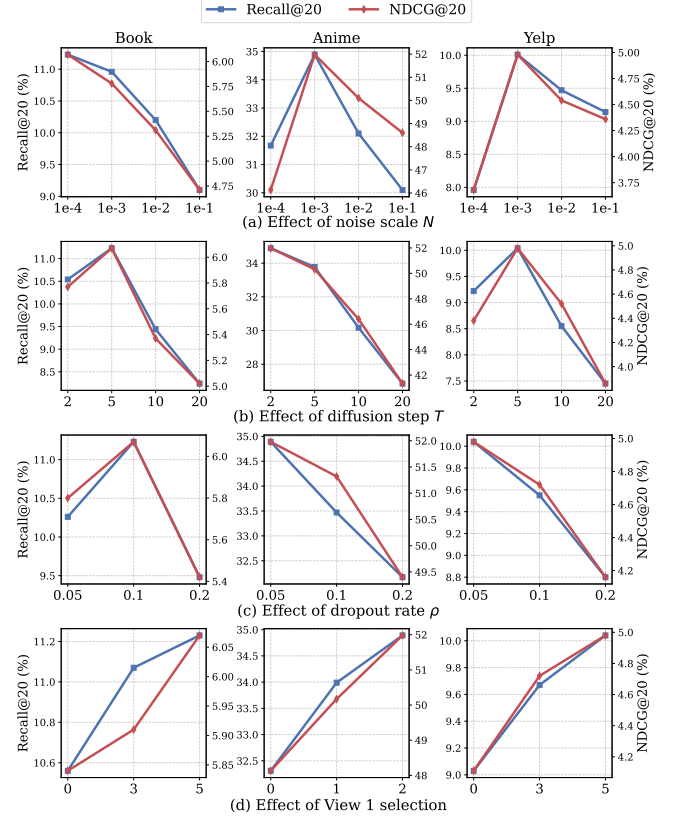
5.1.4 Implementation details. All models are implemented in PyTorch and trained on a single NVIDIA V100 GPU (32GB). For baseline methods, we use the best hyperparameters reported in their original papers or obtained via validation set tuning. For CCL-Diff, we build upon the diffusion framework of DiffRec [33] and CF-Diff [10]. The noise scale N , the noise lower bound $\alpha_{\min}$, and the noise upper bound $\alpha_{\max}$ are searched in $\{1e-4, 1e-3, 1e-2, 1e-1\}$, $\{5e-4, 1e-3, 5e-3\}$, and $\{5e-3, 1e-2\}$, respectively. The diffusion step T and the inference step T' are tuned in $\{2, 5, 10, 20\}$ and $\{0, \frac{T}{4}, \frac{T}{2}\}$, respectively. The step t for View 1 selection is search in $\{0, \lceil \frac{T}{2} \rceil, T\}$. The timestep embedding dimension is fixed at 64, and the dropout ratio ρ is searched in $\{0.05, 0.1, 0.2\}$.

5.2 Experiment Results and Analysis

5.2.1 Performance Comparison with Baselines (RQ1). Table 2 summarizes the performance comparison between CCL-Diff and baseline methods, highlighting two key findings:

(1) Superiority over GNN-based Models. CCL-Diff outperforms all graph-based models across datasets. On the Book dataset, it improves over the strongest GCL baseline, NCL, by **40.6%** in R@10 and **37.9%** in N@10. These substantial gains indicate that CCL captures more informative and generalizable user-item representations beyond what is achievable through traditional graph contrastive learning, which often relies on handcrafted augmentations.

(2) Advantages over Diffusion-based Models. CCL-Diff also consistently exceeds the performance of existing diffusion-based recommenders. On the Anime dataset, it improves by **9.7%** in R@10 and **8.5%** in N@10 over CF-Diff. On Yelp, it achieves gains of 4.4%

**Figure 3: Effect of hyperparameters N , T , ρ , and View 1.**

and 3.0%, respectively. These results confirm that CCL-Diff enhances diffusion models’ representation capacity, making it effective across domains with varying sparsity and interaction patterns.

5.2.2 Hyperparameter Sensitivity Analysis (RQ2). We evaluate the sensitivity of CCL-Diff through ablation tests on key hyperparameters. Figure 3 shows the performance on the validation set for each hyperparameter, with other parameters fixed.

Noise Scale (N): This controls the magnitude of Gaussian noise injected during the forward diffusion process. The best performance occurs with smaller values ($N = 1e-3$ for Anime and Yelp, and $N = 1e-4$ for Book), indicating that mild noise perturbation is effective for contrastive learning without overwhelming the semantic content of the original interactions.

Diffusion Step (T): This controls the depth of the forward diffusion process. Performance improves with smaller T , peaking at $T = 5$ for Book and Yelp, and $T = 2$ for Anime. These results suggest that shallow diffusion is often sufficient due to the inherent sparsity of user-item interactions in recommendation data.

Dropout Ratio (ρ): This parameter is explored for both node and edge dropout. The optimal setting is $\rho = 0.1$ for Book and Yelp, and $\rho = 0.05$ for Anime. These results demonstrate that moderate perturbations enhance model robustness, while excessive dropout may degrade the structural integrity of the interaction graph.

Selection of View 1: We further ablate the choice of View 1 in the CCL objective, comparing three candidates: the clean input $\mathbf{x}_0$, an

⁴<https://www.yelp.com/dataset/>

Table 2: Overall performance comparison between the baselines and CCL-Diff on three datasets. Best results are in bold, second-best underlined. "* implies the improvements are statistically significant (p-value < 0.05) under one-sample t-tests.**

Methods	Book				Anime				Yelp			
	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20
MF	0.0437	0.0689	0.0264	0.0339	0.1319	0.1943	0.3764	0.3631	0.0341	0.0560	0.0210	0.0276
LightGCN	0.0534	0.0822	0.0325	0.0411	0.2071	0.3043	0.3937	0.3824	0.0540	0.0904	0.0325	0.0436
SGL	0.0481	0.0789	0.0301	0.0377	0.1994	0.2918	0.3748	0.3652	0.0503	0.0896	0.0311	0.0424
NCL	0.0559	0.0841	0.0367	0.0439	0.2063	0.3047	0.3915	0.3819	0.0533	0.0883	0.0318	0.0430
XSimGCL	0.0547	0.0833	0.0346	0.0423	0.1764	0.2679	0.3361	0.3273	0.0559	0.0936	0.0346	0.0447
CDAE	0.0538	0.0737	0.0361	0.0422	0.2036	0.2994	0.4067	0.4001	0.0444	0.0703	0.0280	0.0360
MultiDAE	0.0571	0.0855	0.0357	0.0442	0.2142	0.3085	0.4177	0.4125	0.0522	0.0864	0.0316	0.0419
MultiVAE	0.0628	0.0935	0.0393	0.0485	0.2167	0.3106	0.4397	0.4310	0.0567	0.0945	0.0344	0.0458
DiffRec	<u>0.0695</u>	<u>0.1010</u>	<u>0.0451</u>	<u>0.0547</u>	0.2104	0.3012	0.5047	0.4649	0.0581	0.0960	0.0363	0.0478
G-Diff	0.0601	0.0913	0.0389	0.0494	0.1461	0.2102	0.3919	0.3531	0.0483	0.0777	0.0304	0.0394
CF-Diff	0.0683	0.0994	0.0437	0.0539	<u>0.2191</u>	<u>0.3155</u>	<u>0.5152</u>	<u>0.4748</u>	<u>0.0585</u>	<u>0.0971</u>	<u>0.0368</u>	<u>0.0483</u>
CCL-Diff	0.0786	0.1123	0.0506	0.0607	0.2403	0.3489	0.5592	0.5198	0.0611	0.1004	0.0379	0.0498
Improv.(%)	13.1*	11.2*	12.2*	11.0*	9.7*	10.6*	8.5*	9.5*	4.4*	3.4*	3.0*	3.1*

Table 3: Ablation study on components in CCL-Diff.

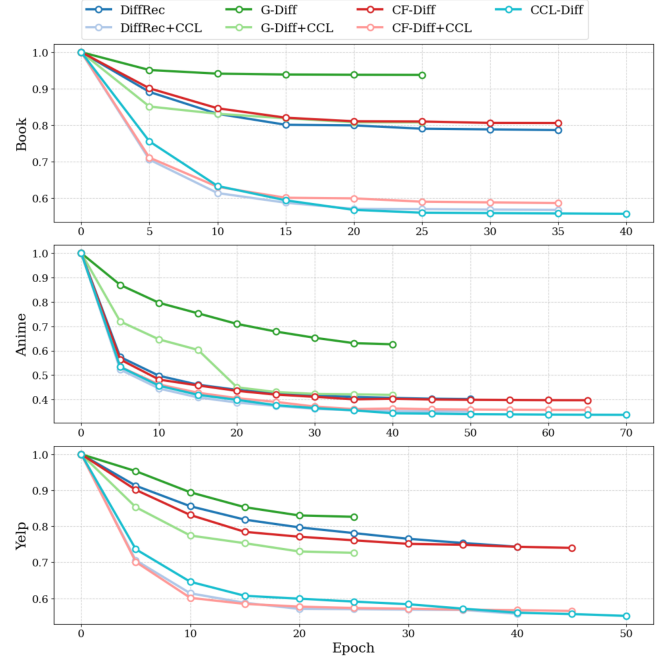
Dataset	Method	R@10	R@20	N@10	N@20
Book	CCL-Diff	0.0786	0.1123	0.0506	0.0607
	w/o CCL	0.0703	0.1017	0.0451	0.0553
	w/o GraphAug.	0.0773	0.1101	0.0487	0.0579
	w/o A-MLP	0.0775	0.1097	0.0493	0.0583
Anime	CCL-Diff	0.2403	0.3489	0.5592	0.5198
	w/o CCL	0.2183	0.3164	0.5213	0.4801
	w/o GraphAug.	0.2365	0.3409	0.5497	0.5103
	w/o A-MLP	0.2363	0.3379	0.5502	0.5121
Yelp	CCL-Diff	0.0611	0.1004	0.0379	0.0498
	w/o CCL	0.0593	0.0973	0.0369	0.0484
	w/o GraphAug.	0.0602	0.0991	0.0373	0.0493
	w/o A-MLP	0.0604	0.0987	0.0372	0.0491

intermediate state $\mathbf{x}_{T/2}$, and the fully noised state $\mathbf{x}_T$. Experimental results consistently favor $\mathbf{x}_T$ across all datasets and metrics. This validates our theoretical analysis in Section 4.5: using the maximally corrupted state $\mathbf{x}_T$ provides the strongest contrastive signal against the denoised output $\hat{\mathbf{x}}_0$, thereby encouraging the model to learn more robust and noise-invariant representations.

5.2.3 Ablation Study on Model Components (RQ3). We conduct an ablation study to evaluate the contribution of each core component in CCL-Diff by systematically removing the following modules: (1) the CCL objective, (2) graph augmentation, and (3) the A-MLP decoder. The results are summarized in Table 3.

Effect of Removing CCL (w/o CCL): Removing CCL results in a noticeable performance drop across all datasets, especially on Anime (e.g., 9.1% decrease in R@10), emphasizing CCL’s role in learning noise-invariant, discriminative user representations.

Effect of Removing Graph Augmentation (w/o GraphAug.): Disabling graph augmentation (node and edge dropout) causes a

**Figure 4: Normalized training loss curves for DMs, DMs+CCL, and CCL-Diff. All losses are scaled by their initial values to facilitate comparison of convergence behavior.**

moderate performance decline. Unlike typical graph contrastive learning methods that use augmentation to generate contrastive views, our method uses augmentation solely to expose the model to structurally perturbed inputs, encouraging robustness in recovering the original interaction graph. This explains the consistent but modest performance drop when augmentation is removed.

Effect of Removing Cross-Attention (w/o A-MLP): Replacing

Table 4: Performance comparison of DMs with CCL.

Methods	Book				Anime				Yelp			
	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20
DiffRec	0.0695	0.1010	0.0451	0.0547	0.2104	0.3012	0.5047	0.4649	0.0581	0.0960	0.0363	0.0478
DiffRec+CCL	0.0767	0.1101	0.0496	0.0592	0.2301	0.3317	0.5369	0.4967	0.0595	0.0973	0.0370	0.0485
Improv.(%)	10.4	9.0	10.0	8.2	9.4	10.1	6.4	6.8	2.4	1.4	1.9	1.5
G-Diff	0.0601	0.0913	0.0389	0.0494	0.1461	0.2102	0.3919	0.3531	0.0483	0.0777	0.0304	0.0394
G-Diff+CCL	0.0671	0.0997	0.0443	0.0546	0.1754	0.2501	0.4331	0.3884	0.0533	0.0891	0.0337	0.0430
Improv.(%)	11.6	9.2	13.9	10.5	20.1	19.0	10.5	10.0	10.4	14.7	10.9	9.1
CF-Diff	0.0683	0.0994	0.0437	0.0539	0.2191	0.3155	0.5152	0.4748	0.0585	0.0971	0.0368	0.0483
CF-Diff+CCL	0.0757	0.1005	0.0490	0.0588	0.2287	0.3296	0.5302	0.4921	0.0599	0.0990	0.0375	0.0492
Improv.(%)	10.8	1.1	12.1	9.1	4.4	4.5	2.9	3.6	2.4	2.0	1.9	1.9

Table 5: Computational efficiency comparison across three datasets. Reported metrics include the number of epochs, training time per epoch (in seconds), total time (in minutes, including training and evaluation), and parameter count (in millions).

Methods	Book				Anime				Yelp			
	Epoch	PerEp.	Total	Param.	Epoch	PerEp.	Total	Param.	Epoch	PerEp.	Total	Param.
DiffRec	35	36.8	32.37	190.00	50	7.9	11.77	13.96	40	11.6	12.08	68.84
DiffRec+CCL	35	36.1	29.77	190.00	50	7.2	10.08	13.96	40	11.1	10.57	68.84
G-Diff	25	737.5	610.17	190.22	40	16.8	21.32	14.03	25	210.9	174.72	69.31
G-Diff+CCL	25	721.6	564.32	190.22	40	16.2	20.02	14.03	25	203.7	153.62	69.31
CF-Diff	40	70.6	58.23	186.63	65	17.7	29.78	14.10	45	24.9	26.58	67.88
CF-Diff+CCL	40	68.3	51.30	186.63	65	17.1	25.85	14.10	45	23.7	23.75	67.88
CCL-Diff	40	35.7	36.78	196.16	70	7.1	13.42	14.48	50	10.7	13.53	71.11

A-MLP with simple feature concatenation results in a clear performance reduction, particularly on Anime (e.g., 3.2% decrease in R@20). This highlights A-MLP’s importance in capturing temporal dynamics through adaptive feature fusion.

5.2.4 Generalizability of CCL to Existing DMs (RQ4). We integrate CCL into three existing diffusion-based recommenders: DiffRec, G-Diff, and CF-Diff, as shown in Table 4. CCL consistently improves performance across all models and datasets. Specifically, it boosts DiffRec’s Recall@10 by 10.4% on Amazon-Book and G-Diff’s Recall@10 by 20.1% on Anime. Even CF-Diff, which already uses advanced collaborative signals, benefits from CCL, confirming its broad applicability and complementary value in denoising-based recommendation systems.

5.2.5 Computational Efficiency Analysis (RQ5). We evaluate the computational efficiency of CCL-Diff from two angles: optimization efficiency (measured by loss convergence dynamics in Figure 4) and resource cost (including parameter size and running time in Table 5). Here, the reported “PerEp.” time strictly refers to the training time per epoch, while the “Total” time includes both training and evaluation.

A key observation is that integrating CCL into existing DMs (DiffRec, G-Diff, CF-Diff) adds **zero extra parameters** while reducing total runtime by 6.1%-14.4%, primarily due to faster convergence, as shown in Figure 4. For example, DiffRec+CCL reduces total time by 8% on Book (29.77 vs. 32.37 minutes) without any

increase in model size. The standalone CCL-Diff framework, despite a modest 3% increase in parameters compared to DiffRec on Book, achieves significant speedups: it runs 19.7× faster per epoch than G-Diff and completes training 2.0× faster than CF-Diff on Yelp, while maintaining efficiency comparable to DiffRec. These efficiency gains are contributed to CCL’s intrinsic regularization effect, which accelerates convergence within the same epoch budget (Figure 4), despite the slight increase in model capacity.

6 Conclusion

This work introduces CCL-Diff, a novel diffusion-based recommendation framework that addresses the lack of consistency in representations across diffusion timesteps in existing models. By leveraging CCL, which leverages the intrinsic multi-view nature of the diffusion process to align denoised embeddings from different noise levels, we eliminate the need for handcrafted data augmentations. The A-MLP module captures temporal dynamics through cross-attention, adaptively fusing user features with step-specific embeddings. Extensive experiments on three real-world benchmarks demonstrate that CCL-Diff consistently outperforms state-of-the-art models by up to 13.1% in R@10. Notably, CCL serves as a plug-and-play component that enhances existing diffusion recommenders in both accuracy and convergence speed, without adding computational overhead, making it an effective solution for sparse and noisy recommendation data.

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